

AI-Powered Interactive Analysis of Cryptocurrency and Stock Markets Using Machine Learning Techniques for Risk and Trend Evaluation

ISBM 2026 | Paper ID: 11

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Research Motivation

Why analyze cryptocurrency and stock markets together?

Cryptocurrency markets

- Operate continuously (24/7)
- Experience rapid price fluctuations
- Show higher volatility and risk

Traditional stock markets

- Represent established financial systems
- Include major indices such as NASDAQ and S&P 500
- Follow regulated trading schedules

A unified analytical framework is needed to evaluate:

Market trends

Volatility behaviour

Risk patterns

Market conditions

Research gap

Previous studies mainly focused on individual aspects:

- Cryptocurrency volatility analysis
- Crypto-stock correlation
- Market forecasting

Proposed Approach

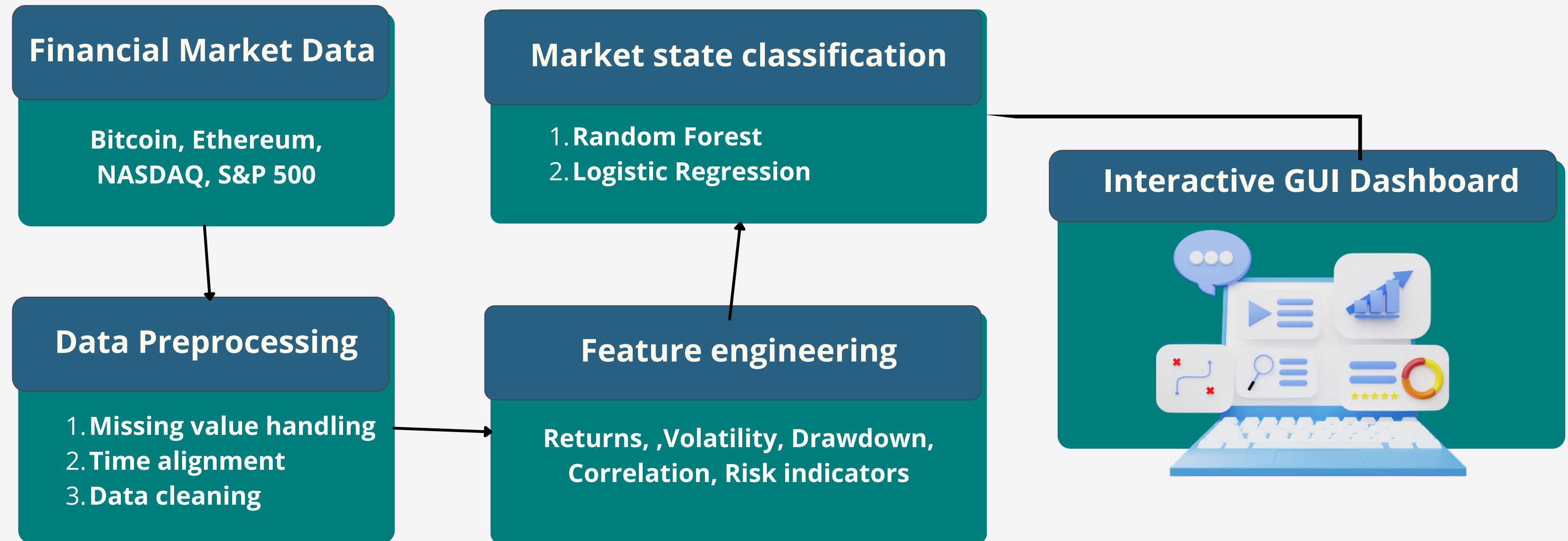
Data preprocessing

Financial feature engineering

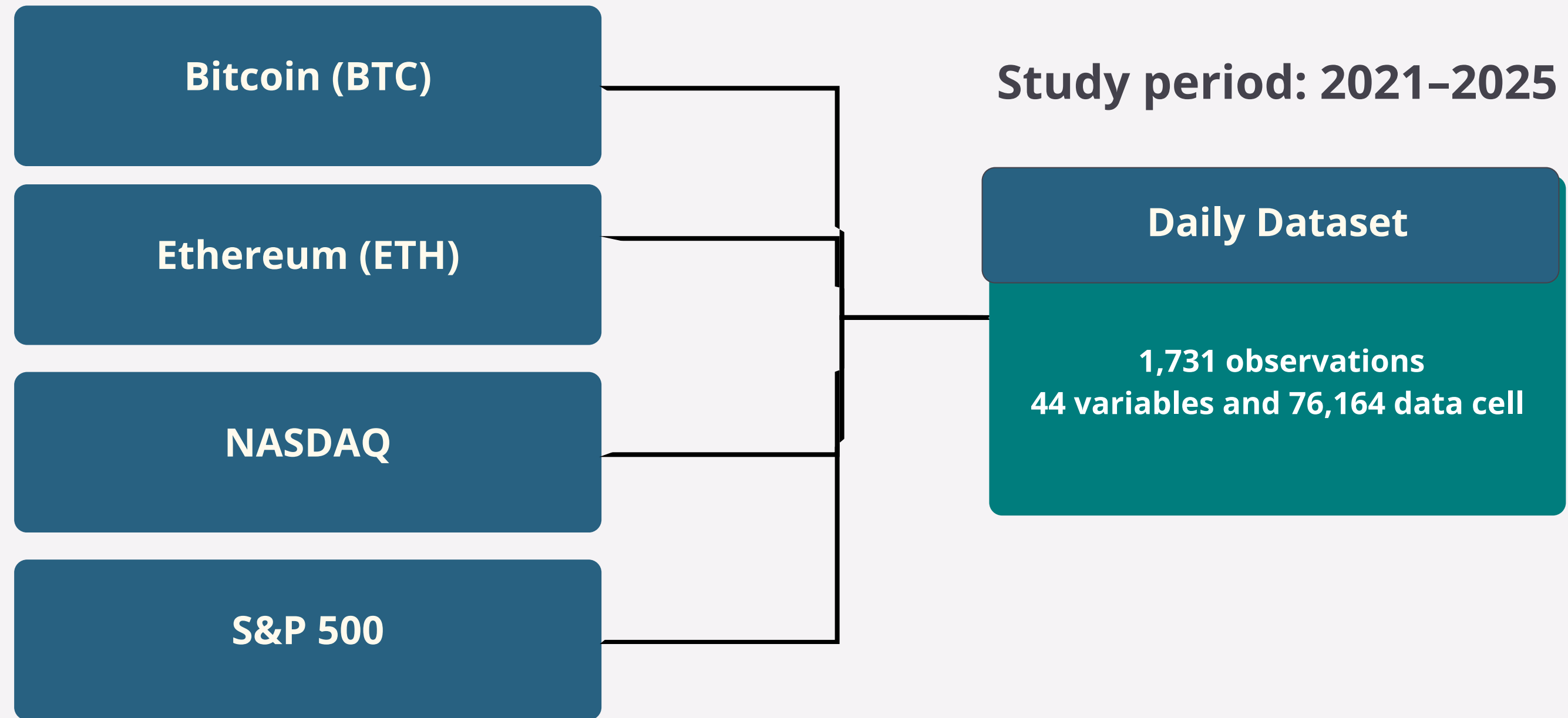
Machine learning classification

Interactive visualization system

Proposed Framework



Dataset



Feature Engineering



Returns



Rolling volatility



Drawdown



Rolling correlation



Liquidity-cost proxy



Risk ratio



Boom flag

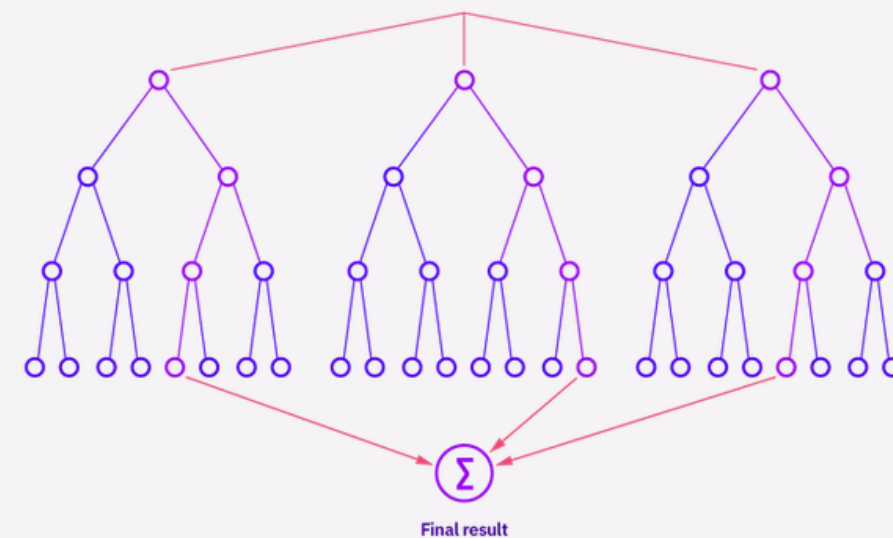


Crash flag

Machine Learning Models

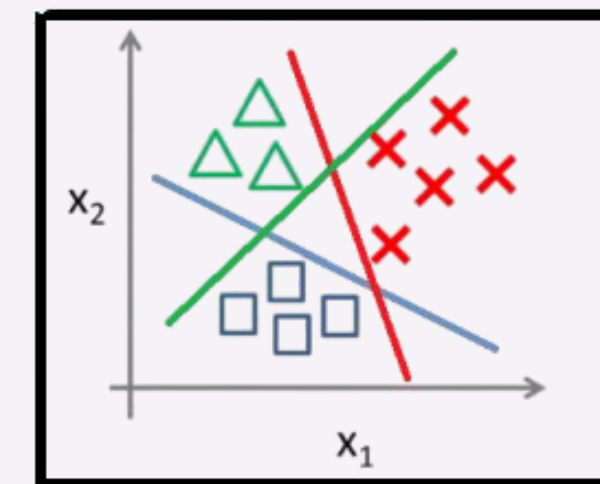
Random Forest

1. **Handles nonlinear relationships**
2. **Robust to high-dimensional data**
3. **Effective for multiclass**



Logistic Regression

1. **Highly interpretable model**
2. **Efficient baseline classifier**
3. **Suitable for multiclass prediction**



Results

Model	Accuracy	Weighted F1-score
Random Forest	99.50%	99.50%
Logistic Regression	97.70%	98.00%

Interactive Market Intelligence Dashboard

Data Workspace

Select analysis layer

Daily Analysis

Upload a custom CSV file optional

Drag and drop file here

Limit 200MB per file • CSV

Browse files

Preprocessing Strategy

Select missing-value treatment

Interpolate numer...

Recommended option: Interpolate numeric + fill remaining. This is appropriate for continuous financial time-series data and preserves the dataset size.

MarketPulse: Crypto vs Stock Market Intelligence

An interactive analytics dashboard for trends, volatility, risk, and market-condition prediction from 2021 to 2025.

InputPreprocessModelVisual InsightsInteractive Output

Active analysis layer: Daily Analysis | Preprocessing strategy: Interpolate numeric + fill remaining

01 Data Overview02 Data Preparation03 Interactive Model04 Visual Insights

Data Overview: Dataset Profile and Completeness

TOTAL ROWS1,731

TOTAL ATTRIBUTES44

FIRST OBSERVATION2021-02-03

LAST OBSERVATION2025-10-30

Dataset Completeness Snapshot

TOTAL DATA POINTS76,164

AVAILABLE DATA POINTS71,922

MISSING DATA POINTS4,242

MISSING DATA RATE5.57%

Raw Data Preview

	datetime	btc_open	btc_high	btc_low	btc_close	btc_volume	eth_open	eth_high	eth_low	eth_close	eth_volume	nasdaq_open	nasdaq_high	nasdaq_low	nasdaq_close	nasdaq_volume	sp500_close	btc_return	btc_return_30d	btc_volatility_30d	btc_drawdown	btc_crash_flag	btc_boom_flag
0	2021-02-03 00:00:00	35472.71	37662.63	35362.38	37618.87	80784.3337	None	None	None	None	None	13513.4	13596.1	13400.3	13427.2	0	3830.17	0.0607	0.176	0.0547	-0.073	0	1
1	2021-02-04 00:00:00	37620.26	38708.27	36161.95	36936.66	92080.7359	None	None	None	None	None	13442.8	13579	13356.6	13571.7	0	3871.74	-0.0181	0.088	0.0539	-0.0898	0	0
2	2021-02-05 00:00:00	36936.65	38310.12	36570	38290.24	66681.3343	None	None	None	None	None	13563.3	13646.4	13527.9	13611.8	0	3886.83	0.0366	0.0414	0.0522	-0.0565	0	0
3	2021-02-06 00:00:00	38289.32	40955.51	38215.94	39186.94	98757.3112	None	None	None	None	None	13563.3	13646.4	13527.9	13611.8	0	3886.83	0.0234	-0.0062	0.0507	-0.0344	0	0
4	2021-02-07 00:00:00	39181.01	39700	37351	38795.69	84363.6798	None	None	None	None	None	13563.3	13646.4	13527.9	13611.8	0	3886.83	-0.01	-0.044	0.0504	-0.044	0	0
5	2021-02-08 00:00:00	38795.69	46794.45	37988.89	46374.87	138597.5369	None	None	None	None	None	13627.7	13699.9	13604.2	13693.1	0	3915.59	0.1954	0.1568	0.0617	0	0	1
6	2021-02-09 00:00:00	46374.86	48142.19	44961.09	46420.42	115499.8617	None	None	None	None	None	13687.2	13742.5	13638.9	13705.2	0	3911.23	0.001	0.2168	0.0609	0	0	0
7	2021-02-10 00:00:00	46420.42	47310	43727	44807.58	97154.1822	None	None	None	None	None	13700.2	13776.8	13525.6	13672.9	0	3909.88	-0.0347	0.2656	0.0595	-0.0347	0	0
8	2021-02-11 00:00:00	44807.58	48678.9	43994.02	47969.51	89561.0815	None	None	None	None	None	13664.9	13750.5	13620.2	13717.9	0	3916.38	0.0706	0.4087	0.0598	0	0	1
9	2021-02-12 00:00:00	47968.66	48985.8	46125	47287.6	85870.0357	None	None	None	None	None	13718.2	13827.6	13653.7	13826.4	0	3934.83	-0.0142	0.2653	0.0579	-0.0142	0	0

Interactive Market Intelligence Dashboard

The GUI allows users to:

- 1. Select a historical date
- 2. Adjust market indicators using sliders
- 3. Click "Predict Market Condition"
- 4. View predicted output

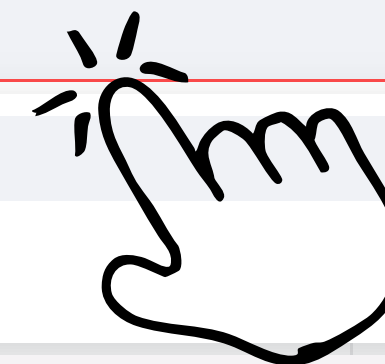
1

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2

2025-10-23

2025-10-24

2025-10-25

2025-10-26

2025-10-27

2025-10-28

2025-10-29

2025-10-30

2025-10-30



Interactive Market Intelligence Dashboard

3

Tune Market Indicators

Adjust the sliders to simulate changes in returns, volatility, drawdown, correlation, and risk. Then generate a fresh market-state prediction.



Interactive Market Intelligence Dashboard

Scenario Input Submitted to the Model

	btc_return	btc_return_30d	btc_volatility_30d	btc_drawdown	btc_convexity	btc_liquidity_cost	eth_return	eth_return_30d	eth_volatility_30d	eth_drawdown	eth_convexity	eth_liquidity_cost	nasdaq_return	nasdaq_return_30d	nasdaq_volatility_30d	nasdaq
0	-0.0157	-0.0222	0.0234	-0.1159	0.0082	51566.4293	0	0	0	-0.4937	0.0172	11068906.0969	0	-0.0024	0.0004	

Generate Market Condition Output

Market Condition Output

PREDICTED CLASS ID

0

PREDICTED MARKET STATE

Normal

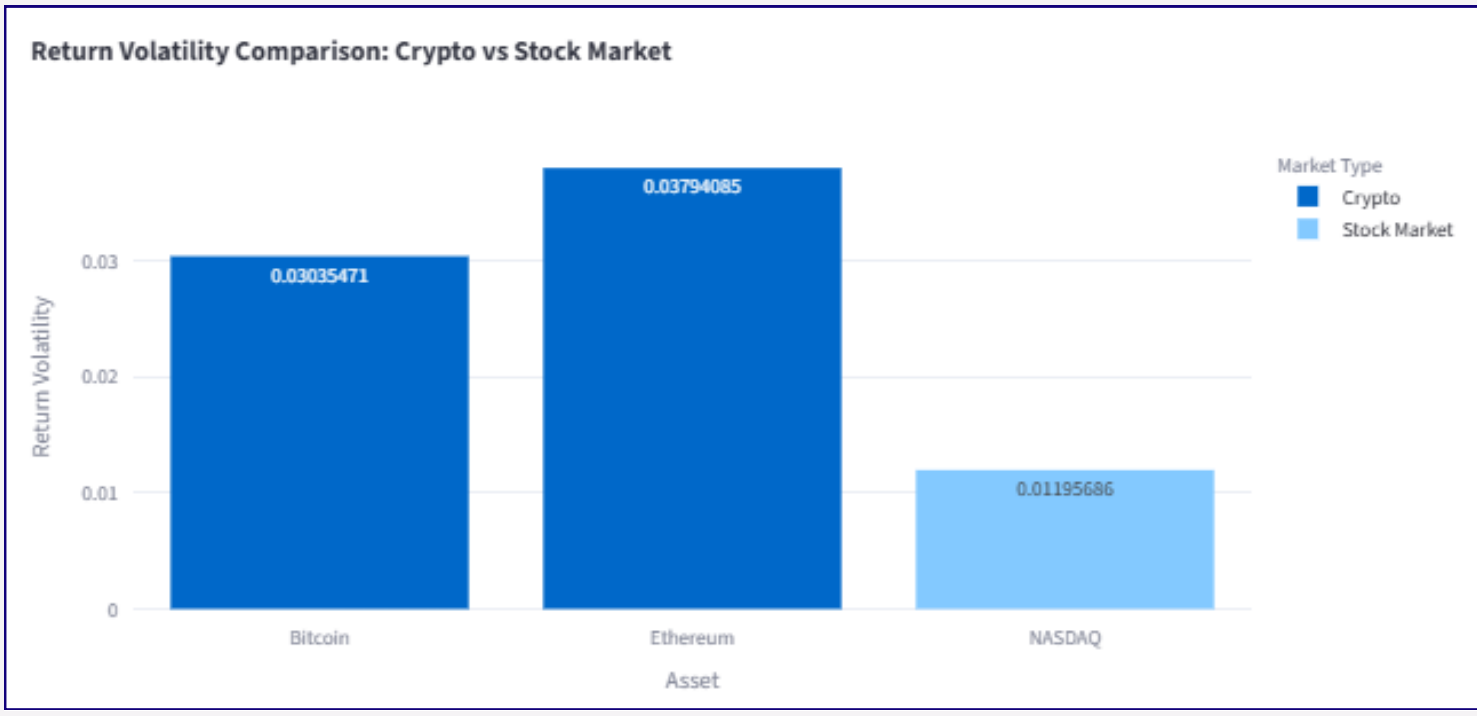
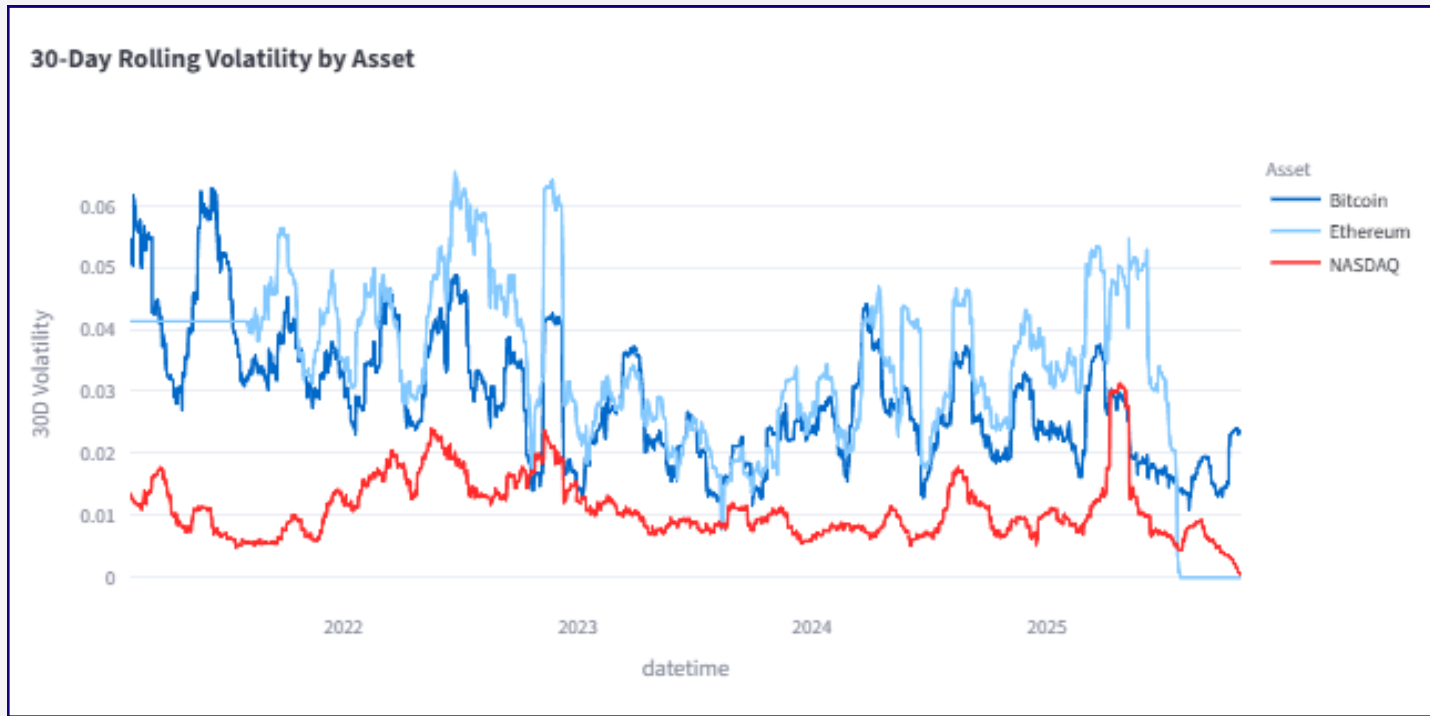
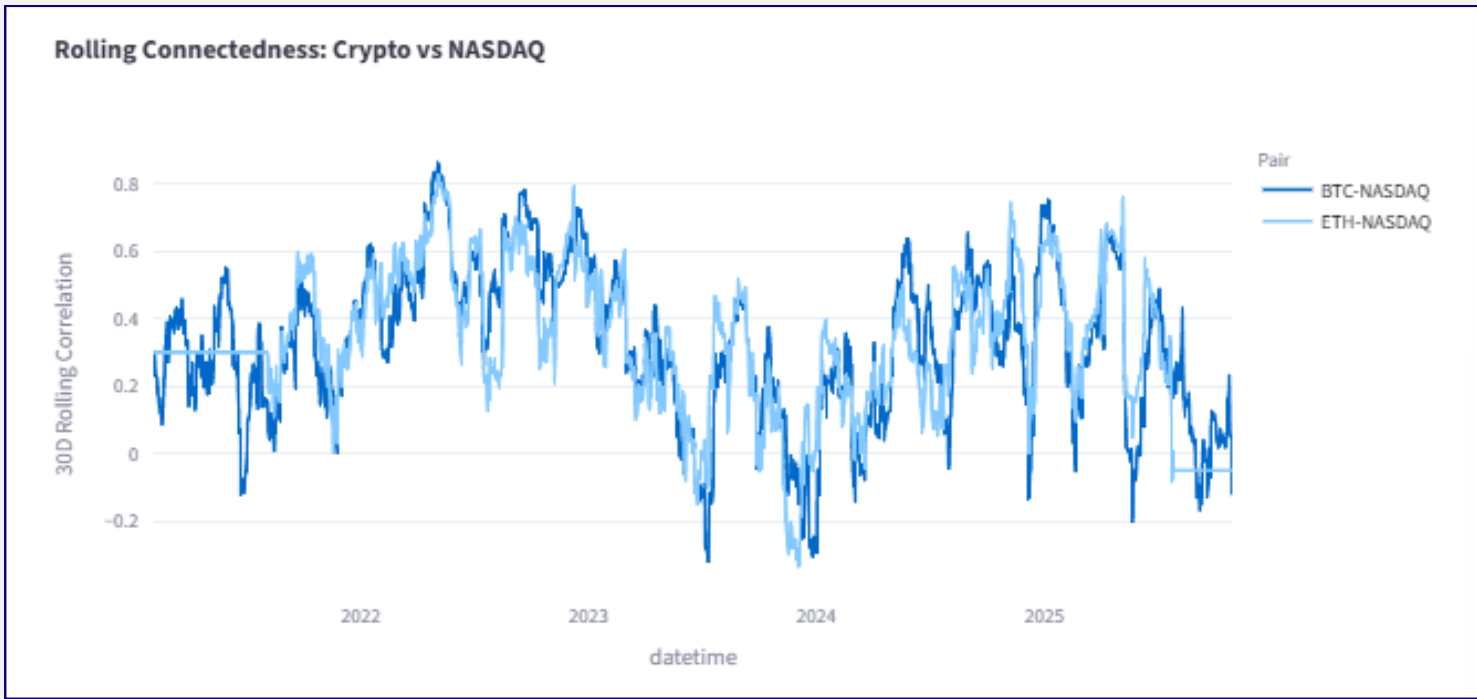
MODEL CONFIDENCE

1.000

Market Status: NORMAL condition detected. The selected scenario does not show strong boom or crash behavior.

4

Market Analysis Output



Key findings

Cryptocurrency behaviour

BTC and ETH showed stronger fluctuations compared with stock indices.

Risk analysis

Cryptocurrency assets demonstrated higher volatility and risk variation.

Market relationship

Machine learning models successfully classified historical market states.

Conclusion

1. Developed an interactive crypto-stock analysis system
2. Combined financial indicators with machine learning classification
3. Random Forest achieved 99.5% accuracy

Future Work

1. Real-time financial data integration
2. Additional financial assets
3. Macroeconomic variables
4. Explainable AI approaches

Thank You
Questions?

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